专业术语表

Technical Glossary

电缆局部放电检测软件 · 中英对照 / Cable PD Testing Software · Chinese–English

本词表收录客户在电缆试验、校准、数据查看与报告中常见的 80 项术语。中文名称、英文名称、缩写或符号及使用说明对应排列。

This glossary contains 80 terms used in cable testing, calibration, data review and reporting. Chinese and English names, abbreviations or symbols, and usage notes are aligned.

缩写栏中的“—”表示本词表不指定缩写。术语解释不规定试验电压、合格阈值或设备能力；这些内容应按交付设备文件、批准的试验方案和现场规程确认。

A dash means no abbreviation is assigned here. Definitions do not prescribe test voltage, acceptance limits or equipment capabilities; use the delivered equipment documentation, approved test procedure and site rules.

01 电压与试验 / Voltage and testing

编号 / 符号 No. / Symbol	术语 / Term	含义与使用说明 / Meaning and usage
01 PD	**局部放电** Partial discharge	发生在绝缘局部、未贯穿导体之间全部绝缘的放电。评价须结合原始信号、背景干扰及试验条件。 A localized discharge that does not bridge the entire insulation between conductors. Assess it with original signals, interference and test conditions.
02 DAC	**阻尼交流；振荡波试验电压** Damped alternating current	幅值随时间衰减的交流振荡试验电压。[2] 本词表中 DAC 不表示数模转换器。 An oscillating AC test voltage whose amplitude decays with time. [2] DAC here does not mean a digital-to-analogue converter.
03 OWTS	**振荡波试验系统** Oscillating wave test system	用于产生振荡波试验电压并进行测量的系统名称。具体能力以交付设备文件为准。 A system that produces oscillating-wave test voltage and performs measurements. Capabilities are defined by the delivered equipment documentation.

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04 VLF	**超低频** Very low frequency	电缆试验语境中的低频交流试验方式。[2] 记录中应注明具体波形和频率，不仅写 VLF。 A low-frequency AC cable-test method. [2] Record the actual waveform and frequency, not merely VLF.
05 —	**电压试验** Voltage test	软件中的加压操作，不是机械压力试验。 The application’s voltage operation, not a mechanical pressure test.
06 —	**耐压试验** Withstand voltage test	按批准的试验方案施加规定电压；不把所有加压测量都称为耐压。 Application of prescribed voltage under an approved procedure; not every voltage measurement is a withstand test.
07 U0	**额定对地电压** Rated conductor-to-earth voltage	电缆额定电压 U0/U 中导体与地或金属屏蔽之间的额定电压，通常以有效值表示。与额定相间电压 U 区分。 The rated voltage between a conductor and earth or metallic screen in a cable rating U0/U, normally expressed as RMS. Distinguish it from U.
08 Peak / pk	**峰值** Peak value	波形相对于零参考的最大幅值。核对显示是正峰值、负峰值还是绝对峰值；峰值不是有效值。 The maximum waveform amplitude relative to zero. Check whether a value is a positive, negative or absolute peak; peak is not RMS.
09 RMS	**有效值；均方根值** Root mean square value	在规定时间区间内对波形平方取平均后再开平方得到的数值。非正弦或衰减波形不能无条件按正弦关系换算。 The square root of the mean squared waveform over a specified interval. Sinusoidal conversion factors do not apply unconditionally to non-sinusoidal or decaying waveforms.
10 —	**目标电压** Target voltage	本次试验计划或自定义设置要求达到的电压，不是设备已经输出或测得的电压。 The voltage requested by the test plan or custom setting, not proof of an output or measured voltage.

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11 —	**电压反馈** Voltage feedback	设备返回的电压测量值。核对测量来源、单位和当前有效性；失效或缺失值不按零解释。 A voltage measurement returned by the equipment. Check its source, units and current validity; an invalid or missing value is not zero.
12 HV	**高电压** High voltage	电压等级用语。等级边界依适用标准和应用而定；不能据此确定本次试验电压。 A voltage-class term. Class boundaries depend on the applicable standard and application; the term does not specify the test voltage.
13 MV	**中电压** Medium voltage	配电系统常用电压等级用语。不同地区的范围规定可能不同。 A voltage-class term commonly used for distribution systems. Its range may differ between jurisdictions.
14 AC	**交流** Alternating current	随时间交替改变方向的电流；AC voltage 表示交流电压。 Current that alternates in direction with time; AC voltage means alternating voltage.
15 DC	**直流** Direct current	方向不交替改变的电流；DC voltage 表示直流电压。不得与交流试验方式混同。 Current that does not alternate in direction; DC voltage means direct voltage. Distinguish DC testing from AC testing.
16 —	**工频** Power frequency	电力系统运行频率，通常为 50 Hz 或 60 Hz。实际数值以当地系统为准。 The operating frequency of a power system, commonly 50 Hz or 60 Hz. Use the frequency of the applicable system.
17 p-p	**峰峰值** Peak-to-peak value	波形最大值与最小值之差。与单侧峰值区分，不能直接用峰峰值作为试验电压设定。 The difference between waveform maximum and minimum. It is not a one-sided peak and must not be substituted directly for the test-voltage setting.

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18 U	**额定相间电压** Rated voltage between conductors	电缆额定电压 U0/U 中导体之间的额定电压。不得直接替代以 U0 为基准的倍数设置。 The rated voltage between conductors in a cable rating U0/U. Do not substitute it for U0 in a voltage-multiplier setting.
19 —	**分级加压计划** Stepped-voltage plan	由多个目标电压级别及对应试验次数组成的计划。计划数值不等于实际完成的试验记录。 A plan containing voltage levels and associated test counts. Planned values are not records of tests actually completed.

02 试验回路与电缆 / Test circuit and cable

编号 / 符号 No. / Symbol	术语 / Term	含义与使用说明 / Meaning and usage
20 HV source	**高压电源** High-voltage source	为试验回路提供高电压的装置。连接成功不代表输出已关闭或回路已安全。 Equipment supplying high voltage to the test circuit. A successful connection does not confirm that output is off or the circuit is safe.
21 DUT	**试品；被试设备** Test object / Device under test	接受试验的电缆或设备。DUT 是 Device under test 的缩写。 The cable or equipment under test. DUT expands to Device under test.
22 —	**电抗器** Reactor	试验回路中的电感设备，可参与形成振荡或谐振回路。具体作用按设备接线图确认。 An inductive device that may form part of an oscillating or resonant test circuit. Confirm its function from the equipment circuit diagram.
23 Ck	**耦合电容器** Coupling capacitor	在测量回路中耦合局放信号的电容器。Ck 是常见回路符号，不是通用设备型号。 A capacitor used to couple PD signals in a measuring circuit. Ck is a common circuit symbol, not a universal equipment model.

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24 —	**分压器** Voltage divider	将被测高电压按规定比例转换为可测信号的装置。比例、连接及额定参数以设备文件为准。 A device that scales high voltage to a measurable signal. Use the ratio, connections and ratings specified in its documentation.
25 —	**电缆线芯** Cable core	电缆中的导体及其相关绝缘结构。试验相别必须与实际接线、资产编号一致。 A cable conductor and its associated insulation structure. The test phase must match the physical connections and asset identification.
26 —	**金属屏蔽** Metallic screen / shield	电缆的导电屏蔽结构。其接地和连接方式应按现场批准的试验接线执行。 The conductive screening structure of a cable. Its earthing and connections must follow the approved test arrangement.
27 XLPE	**交联聚乙烯** Cross-linked polyethylene	一种电缆绝缘材料。材料名称本身不说明电缆绝缘当前状态。 A cable-insulation material. Its name alone does not indicate the present condition of the insulation.
28 —	**电缆电容** Cable capacitance	在指定导体与屏蔽或其他导体之间测得的电容。记录时须明确测量对象和单位。 Capacitance measured between a specified conductor and screen or another conductor. Identify the measurement pair and units.
29 L	**电缆长度** Cable length	项目中的总长度或经测量核对的长度。应注明来源和参考端，不把屏幕估算值当作资产定值。 The project cable length or a verified measured length. Record its source and reference end; an on-screen estimate is not an asset-certified value.

03 安全与状态 / Safety and status

编号 / 符号 No. / Symbol	术语 / Term	含义与使用说明 / Meaning and usage
30 E-stop	**急停** Emergency stop	危险情况下用于紧急停止的功能或装置。软件提示不替代设备急停装置和现场安全处置。 A function or device used to stop in an emergency. Software messages do not replace the equipment emergency-stop device or site safety action.
31 —	**急停锁存** Latched emergency stop	急停发生后保留的禁止启动状态。解除急停不应自动恢复高压；重新开始前须完成安全检查。 An inhibited-start state retained after an emergency stop. Release must not automatically restore HV; complete safety checks before restarting.
32 —	**状态未知** Unknown state	软件没有当前有效依据确认设备状态。不得沿用上一正常状态继续试验。 The application has no current valid evidence of equipment state. Do not continue a test on the basis of the previous normal state.
33 Stop	**正常停止** Normal stop	请求受控结束当前任务。健康连接可以按软件设计保留；停止请求或按钮复位均不证明试品已放电。 A request to end the current task in a controlled manner. A healthy connection may be retained; the request or a reset button is not proof of discharge.
34 —	**断开连接** Disconnect	结束软件与设备的通信连接。断开连接不等于输出已停止，也不等于电缆已放电接地。 Ending the communication connection to the equipment. Disconnection does not confirm stopped output or a discharged and earthed cable.
35 —	**资源释放** Resource release	清理采集任务及缓冲区等占用。释放失败时保留异常记录，不反复点击开始。 Releasing acquisition tasks, buffers and related resources. Retain failure details and do not repeatedly click Start after release fails.

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36 —	**离线模式** Offline mode	用于项目管理和已有数据查看分析的工作模式，不进行实机输出。 A mode for project management and review or analysis of existing data, without live equipment output.
37 —	**仿真** Simulation	用于演示、培训或验证流程的模拟运行。仿真记录必须与实测记录区分，不作为验收数据。 Simulated operation for demonstration, training or workflow checking. Keep simulated records separate from measurements; do not use them for acceptance.
38 —	**放电接地** Discharge and earthing	按设备说明和现场规程消除残余电荷并建立接地的现场操作。显示 0 kV、停止或断开均不能替代安全确认。 Site operations to remove residual charge and establish earthing under equipment and site procedures. A 0 kV display, Stop or Disconnect is not safety confirmation.

04 校准与采集 / Calibration and acquisition

编号 / 符号 No. / Symbol	术语 / Term	含义与使用说明 / Meaning and usage
39 q	**视在电荷** Apparent charge	在指定回路中，从试品端子短时注入后能产生与该局放脉冲相同仪器读数的等效电荷。[1] 不等于缺陷处实际电荷。 Equivalent charge which, injected briefly at the test-object terminals in a specified circuit, produces the same instrument reading as the PD pulse. [1] It is not the actual local charge.
40 —	**已知注入电量** Known injected charge	校准器提供的参考电荷量。核对校准器证书、实际档位、极性和单位后录入。 Reference charge supplied by the calibrator. Check its certificate, actual charge level, polarity and units before entry.

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41 —	**校准系数** Calibration coefficient	将测量响应换算为电荷量的系数，适用于对应试品、接线和测量条件。[1] 不把系数简单套用于不同配置。 A factor relating measuring-system response to charge, valid for the associated test object, setup and measurement conditions. [1] Do not transfer it indiscriminately to another configuration.
42 —	**校准档位** Calibration level	校准器的参考电量级别或其对应记录。选择档位并不代表已经完成采集、复核和保存。 A calibrator reference-charge level or its associated record. Selecting a level does not establish that capture, review and saving are complete.
43 —	**量程** Measurement range	采集硬件允许的输入测量范围。改变量程不同于放大显示图形。 The input measurement range of the acquisition hardware. Changing it is not the same as zooming the display.
44 ADC	**ADC 溢出** ADC overflow	模拟数字转换器输入超出所选范围的状态，可能导致削顶或数据失真；须检查记录有效性。 A condition in which the analogue-to-digital converter input exceeds the selected range, potentially causing clipping or distortion. Check record validity.
45 —	**削顶** Clipping	信号峰部因测量范围限制而被截平。削顶记录不适合用于准确的峰值或电荷量评价。 Flattening of signal peaks by a measurement-range limit. A clipped record is unsuitable for accurate peak or charge assessment.
46 —	**采样率** Sample rate	每秒采集的样本数，通常以 Hz、kHz 或 MHz 表示。读取时间轴时须使用记录本身的采样时间基准。 Samples acquired per second, normally expressed in Hz, kHz or MHz. Use the record’s own sampling timebase when reading its time axis.
47 —	**采集长度** Acquisition duration	一次采集实际覆盖的时间区间。它不等于显示窗口宽度，也不等于最终保存的完整周期数。 The time interval actually covered by a capture. It is not the display-window width or the number of complete cycles ultimately saved.

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48 —	**噪声参考** Noise reference	作为背景比较依据的参考记录或数值，不是电缆自然局放记录，也不是合格判据。 A reference record or value for background comparison. It is not a record of natural cable PD or a pass criterion.
49 —	**完整周期** Complete cycle	由有效起止边界界定的一段完整周期波形。末尾不足一个周期的部分不得计为完整周期。 One complete waveform period with valid start and end boundaries. An incomplete tail must not be counted as a complete cycle.
50 —	**冷却等待** Cooling wait	两次试验之间按设备要求保留的等待阶段。采集完成或按钮恢复不表示等待已结束。 The waiting stage required by the equipment between tests. Capture completion or an enabled button does not mean the wait has ended.
51 PDIV	**局放起始电压** Partial discharge inception voltage	升压过程中首次观察到满足规定检测条件的局放时对应的电压。记录阈值、波形及电压量值定义。 Voltage at which PD meeting specified detection conditions is first observed during an increase in voltage. Record threshold, waveform and voltage definition.
52 PDEV	**局放熄灭电压** Partial discharge extinction voltage	从已观察到局放的较高电压降压时，局放按规定检测条件停止出现所对应的电压。与 PDIV 区分。 Voltage at which observable PD ceases under specified detection conditions as voltage is reduced from a level with PD. Distinguish it from PDIV.
53 —	**局放重复率** PD pulse repetition rate	规定时间内检测到的局放脉冲数量与时间之比。统计前须明确检测条件和有效记录区间。 Detected PD pulses per unit time. Specify detection conditions and the valid record interval used for the count.
54 + / −	**极性** Polarity	信号或校准脉冲的正负方向。极性与接线及通道定义有关，不能仅凭正负判断缺陷类型。 The positive or negative direction of a signal or calibration pulse. Polarity depends on connections and channel convention; it does not alone identify a defect.

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55 —	**背景噪声** Background noise	测量记录中非目标局放来源的背景信号。未检出高于噪声的信号，不能单独证明电缆无局放。 Background signals not originating from target PD activity. Failure to detect a signal above the noise is not, by itself, proof of a PD-free cable.

05 波形与定位 / Waveforms and location

编号 / 符号 No. / Symbol	术语 / Term	含义与使用说明 / Meaning and usage
56 TDR	**时域反射测量** Time-domain reflectometry	利用脉冲传播及反射的时间关系测量或核对电缆长度和位置。须确认参考端、时间基准和传播速度。 Measurement or verification of cable length and position from pulse travel and reflection timing. Confirm reference end, timebase and propagation velocity.
57 —	**入射脉冲** Incident pulse	沿电缆向前传播的脉冲，可作为传播时间测量的参考。应与反射脉冲区分。 A pulse travelling forward along the cable that may provide a timing reference. Distinguish it from a reflected pulse.
58 —	**反射脉冲** Reflected pulse	信号遇到阻抗变化后返回的脉冲。反射可能来自端部、接头或其他变化，须结合接线和电缆资料复核。 A pulse returned by an impedance change. Reflections may arise at ends, joints or other changes; review them against connections and cable information.
59 —	**首次到达时间** First-arrival time	信号首次到达参考测量点的时间。它不一定与最大峰值出现的时间相同。 The time of the first signal arrival at the reference measuring point. It need not coincide with the largest peak.
60 v	**传播速度** Propagation velocity	脉冲在电缆中的传播速度。常用 m/μs；1 m/μs = 10^6 m/s，输入时须核对单位。 Pulse propagation speed in the cable, commonly in m/μs. 1 m/μs = 10^6 m/s; check the units before entry.

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61 —	**电缆接头** Cable joint	连接电缆段的中间附件。位置应根据资产或实测资料填写，并明确定位参考端。 An intermediate accessory connecting cable sections. Enter its position from asset or measured information and identify the reference end.
62 —	**电缆终端** Cable termination	电缆端部用于连接其他设备的附件。终端附件与软件中的定位参考端不是同一个概念。 An accessory at a cable end connecting it to other equipment. A physical termination is not the same concept as a software location-reference end.
63 PRPD	**相位分辨局放图谱** Phase-resolved partial discharge pattern	按电压相位、放电量和次数等信息形成的分布图。需结合原始波形和干扰复核，不能单凭图形判定缺陷。 A distribution of PD information by voltage phase, charge and count. Review it with raw waveforms and interference; the pattern alone does not establish a defect.
64 Q-L	**放电量—位置图** Charge versus position plot	显示放电量随电缆位置的分布。读图前核对相别、记录范围、位置参考及电量单位。 A plot of charge against cable position. Check phase, record scope, position reference and charge units before interpretation.
65 N-L	**次数—位置图** Count versus position plot	显示事件数量随电缆位置的分布。事件数受选定记录及统计范围影响，不等于独立缺陷数量。 A plot of event count against cable position. Counts depend on selected records and statistical scope; they are not the number of separate defects.

06 项目与报告 / Projects and reports

编号 / 符号 No. / Symbol	术语 / Term	含义与使用说明 / Meaning and usage
66 —	**当前项目** Active project	已成功加载并作为当前操作对象的项目。列表中仅选中一行，不代表项目已加载。 The successfully loaded project used for current operations. Merely highlighting a list row does not load a project.
67 ALL / A / B / C	**相别** Phase	记录或显示所对应的电缆相别。ALL 通常为全部相别视图；界面切换相别不会改变现场接线。 The cable phase associated with a record or view. ALL normally denotes an all-phase view; switching a view does not change site wiring.
68 —	**候选事件** Candidate event	进入复核流程的候选结果。候选不等于已经确认为真实局放或已经接受的事件。 A result presented for review. A candidate is not automatically confirmed as genuine PD or accepted as an event.
69 —	**接受事件** Accept event	经人工复核后将事件保留到当前分析结果的操作。接受事件不等于设备绝缘状态已判为合格。 An action retaining a reviewed event in the current analysis. Acceptance of an event does not certify that the insulation passes.
70 —	**分析归档** Analysis archive	保存分析状态及其结果的文件或记录。它不能代替原始项目、采集数据和校准文件的备份。 A saved analysis state and its results. It does not replace backup of the original project, acquisition data and calibration files.
71 —	**报告附件** Report attachment	被选入报告的图表或相关资料。取消选入报告通常只影响报告内容，不删除源记录。 A chart or related material selected for a report. Removing its selection normally changes the report only, not the source record.
72 —	**另存为** Save as	用新路径或名称保存，优先保留既有报告。 Saves at a new path/name, preserving an existing report.

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73 —	**备份** Backup	保留可恢复的完整项目副本。副本应包含必要源数据和关联文件，并至少有一份独立于当前工作磁盘。 A complete, recoverable project copy including necessary source data and associated files. Keep at least one copy independent of the working disk.
74 —	**原始记录** Raw record	保留采集信号及其必要元数据的源记录。分析图、截图和报告不能代替完整原始记录。 The source record containing captured signals and necessary metadata. Analysis plots, screenshots and reports do not replace a complete raw record.

07 单位与符号 / Units and symbols

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75 pC	**皮库仑** Picocoulomb	电荷单位：1 pC = 10^{-12} C。p 小写，C 大写。 Unit of charge: 1 pC = 10^{-12} C. Use lowercase p and uppercase C.
76 nC	**纳库仑** Nanocoulomb	电荷单位：1 nC = 10^{-9} C = 1000 pC。 Unit of charge: 1 nC = 10^{-9} C = 1000 pC.
77 kV	**千伏** Kilovolt	1 kV = 1000 V；核对峰值或有效值。 1 kV = 1000 V; check peak or RMS.
78 μs	**微秒** Microsecond	时间单位：1 μs = 10^{-6} s = 1000 ns。定位时间差与传播速度必须使用一致单位。 Unit of time: 1 μs = 10^{-6} s = 1000 ns. Use consistent units for location time differences and propagation velocity.

编号 / 符号 No. / Symbol	术语 / Term	含义与使用说明 / Meaning and usage
79 NaN	**非数值；无有效结果** Not a Number / No valid result	计算机中的非数值标记；出现时表示该项没有可用的有效数值。不能解释为零，也不表示合格。 A non-numeric computing marker. On a display it means no usable valid numeric result is available; it is neither zero nor a pass.
80 tan δ	**介质损耗因数** Dielectric dissipation factor	介质损耗角的正切，用于表征绝缘损耗，无量纲。核对显示为小数、百分数还是其他标度，并结合试验条件评价。 The tangent of the dielectric loss angle, a dimensionless indicator of insulation loss. Check fractional, percentage or other display scaling and interpret with test conditions.

技术参考 / Technical references

[1] OMICRON — A Guide for Partial Discharge Measurements, Part 2: Measurement according to IEC 60270 (2020).

[2] Megger — Excitation voltage options for PD investigations on MV cables (2021).

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